

Element J

Documentation of external evaluation

Documentation of project evaluation by multiple, demonstrably qualified stakeholders and field experts is presented and is synthesized in a consistently specific, detailed, and thorough way; documentation is sufficient in two or more categories to yield meaningful analysis of that evaluation data; the synthesis of evaluations consistently addresses evaluators' specific questions, concerns, and opinions related to design requirements.

After a thorough review, our project and presentation received full marks from both Tom and Meredith of the My4usa team. They concluded that our problems were clearly defined and that our design met all the necessary requirements specified by the zoo staff. The only point of concern highlighted was the durability of the removable handle. There were worries about the potential for the bear to destroy the handle. To address this issue, they suggested several alternatives, such as using roping or designing holes for the bears to utilize instead.

The questions we received during the review session were as follows:

Question: What are alternative designs to the removable handle?

Analysis: Due to concerns over the durability of a removable style handle, the zoo keepers asked if we had any alternative designs for the handle, as to serve as backups if the handle was deemed ill-fit for the spin drum design. The handle could be deemed ill fitting due to concerns over the digestability of said handle if managed to be removed by the bear, as well as concerns of further breakage due to the initial sewerage of the handle, as the bear might be able to gain entry to parts of the toy not meant to be accessed.

Answer: Both a rope and a coverable hole are excellent alternatives to the removable handle design. These options effectively address the durability concerns we were facing as well as the safety concerns. The rope provides a sturdy and flexible solution that can withstand rough handling by the bears as well as being edible in small amounts. Similarly, a coverable hole offers a durable and tamper-proof option, ensuring that the bears cannot easily destroy or remove it. Implementing these alternatives would enhance the overall robustness of the design, ensuring it remains functional and secure for the zoo's needs.

/3	Design does not reflect any competence with engineering skills & tools.	Design reflects tenuous grasp of engineering skills & tools.	Design reflects emerging grasp of engineering skills & tools.	Design reflects proficiency with engineering skills & tools.
/3	Design is incomplete.	Design is flawed.	Design has the potential to be viable.	Design is viable.
/3	Design does not address stakeholder needs.	Design does not adequately meet primary stakeholder needs.	Design may meet some stakeholder needs.	Design meets all stakeholder needs.
Oral Presentation				
/3	Oral presentation is disorganized or too difficult to follow.	Oral presentation is somewhat logically organized, but may be vague or hard to follow.	Oral presentation is mostly logically organized and generally clear.	Oral presentation is logically organized & very clear.
/3	Oral presentation demonstrates insufficient teamwork skills.	Oral presentation demonstrates emerging teamwork skills.	Oral presentation demonstrates adequate teamwork skills.	Oral presentation demonstrates strong teamwork skills.
Community Considerations				
/3	Solution & presentations demonstrate little to no thought about the client's culture & community; do not take into account any social, environmental, or ethical considerations.	Solution & presentations demonstrate superficial thought about the client's culture & community; take into account few, & not necessarily varied, social, environmental, and/or ethical considerations.	Solution & presentations demonstrate adequate thought about the client's culture & community; take into account some varied social, environmental, & ethical considerations.	Solution & presentations demonstrate complex thought about the client's culture & community; take into account multiple & varied social, environmental, & ethical considerations.
/3	Input from the client is neither present nor valued in the design process.	Some input from the client is present.	Input from the client is present and valued in parts of the design process.	Input from the client is present and valued throughout the design process.

Total: 41 /42

Judge name: Tom W.

Notes:

$$\bar{X} = 40$$



e4usa @ BHS Design Showcase Judging

Meredith

126



Project name:

Toy (Tubes)

Student name:

2	4 - Not present or incomplete	1 - Emerging	2 - Adequate	3 - Exceptional
<i>Engineering design process</i>				
3.1	Little to no evidence of engineering design approach.	Some evidence of engineering design approach.	Adequate evidence of engineering design approach.	Strong evidence of engineering design approach.
3.2	Problem is undefined or cannot be inferred from information provided.	Problem vaguely defined based on some stakeholder needs.	Problem generally defined based on stakeholder needs.	Problem clearly defined based on stakeholder needs.
3.3	Design criteria & constraints are not clearly identified, measurable, nor justified.	Design criteria & constraints are partially identified, measurable, and justified.	Design criteria & constraints are identified and partially measurable and justified.	Design criteria & constraints are clearly identified, measurable, and justified.
3.4	Experimentation data is minimal.	Experimentation data is inconsistently valid.	Experimentation data is generally valid.	Experimentation data is consistently valid.
3.5	Experimentation data is undocumented.	Experimentation data is insufficiently documented.	Experimentation data is adequately documented.	Experimentation data is clearly documented.
3.6	Iteration of project design is not present.	Partial iteration of project design is present.	Iteration of project design is present.	Iteration of project design is present and its value illustrated well.
3.7	Design is missing, unfinished, or plagiarized.	Design shows little evidence of creative/original thought.	Design shows some evidence of creative/original thought.	Design shows evidence of creative/original thought.